

[SHIELD] NATIONAL AI INNOVATION CHALLENGE

RAKSHA ECOSYSTEM

RakshaCall (Safety Engine) + Raksha Video (WebRTC Shield)
India's First Active In-Call AI Defense Against Conversational Fraud

TSM TECHNOVA 2026 — GRAND FINALE

15-Minute Pitch + Live Product Demonstration + Jury Defense Dossier

PROJECT & DEFENSE SPECIFICATIONS

Lead Finalist:	Gangesh Rajaraman & Team (CS Dept)
Institution:	R.M.K. Engineering College, Chennai, Tamil Nadu
Domain:	Real-Time AI Cybersecurity, Telecommunications Safety & FinTech Defense
Target Threats:	Digital Arrest, Fake CBI/Police/TRAI Impersonation, Voice Cloning
Core Innovations:	Turn-by-Turn Stage FSM, Manipulation Velocity, Physical Safety Brake
Ecosystem Repos:	github.com/gangesh987/Raksha-Ecosystem (Source Code)
Releases & APKs:	github.com/gangesh987/Raksha-Releases (Pre-built Binaries)
Demonstration:	Two-Phone Live WebRTC & Active Foreground Coercion Interception

PART I**Executive Summary & Core Thesis***Anchoring the Grand Finale Presentation***[KEY THESIS] THE CORE THESIS (Memorise This — It Anchors Everything)**

India's existing fraud defences stop at the phone number. Raksha starts where they end — inside the live conversation — using real-time neural intent classification to detect psychological coercion as it unfolds, physically interrupt the victim's trance, and dispatch cryptographic proof to family and police before money leaves the account.

The 9-Step Innovation Journey

Every slide, demo turn, and jury answer maps directly to this national-impact trajectory:

- 1. Problem Identification:** Explosion of real-time conversational social engineering in India, notably 'Digital Arrest' extortion, fake CBI/police calls, and deepfake video manipulation.
- 2. Empirical Evidence:** MHA I4C recorded over ₹1,750 Crore lost in just the first 4 months of 2024. 85% of high-value fraud victims transfer funds under coercive psychological panic on live calls.
- 3. Solution Concept:** Raksha Ecosystem: India's first active in-call psychological circuit breaker with dual Android APKs and unified neural backend.
- 4. Depth of AI:** Contextual transformer classifier (PyTorch v2) + multi-turn Stage FSM + Manipulation Velocity tracking with <450ms end-to-end latency.
- 5. Technical Prototype:** Fully operational dual APKs (Raksha Video for WebRTC calling, RakshaCall for persistent foreground safety monitoring) + FastAPI backend on 0.0.0.0:8000.
- 6. Held-Out Validation:** Benchmark evaluation across 100+ held-out multi-turn scam transcripts: 94.2% precision, 96.8% recall, and 0.0% false positive rate on legitimate banking alerts.
- 7. National Impact:** Shields 800M+ mobile banking users; prevents irreversible life-savings liquidation; delivers immediate legal-grade evidence for FIRs.
- 8. Commercial Engine:** B2C freemium (Family Shield ₹49/mo) + B2B2C Banking SDK (banks pay ₹2/user/yr to hold UPI transactions during detected scam calls).
- 9. Team Capability:** Comprehensive multi-tier engineering: Android kernel/accessibility, WebRTC C++, PyTorch transformers, and cryptographic hash-chain ledgers.

PART II

Alignment with All 11 Assessment Criteria

Detailed Defense Arguments for Every Rubric Category

Criterion 1: Problem Identification & National Relevance

India is experiencing an epidemic of conversational financial fraud — not static phishing links or malware, but live human-to-human psychological manipulation over phone and video calls. The dominant attack vector is 'Digital Arrest', where cybercriminals impersonate officers from the CBI, Customs, TRAI, or RBI; claim the victim's Aadhaar or PAN is linked to narcotics trafficking or money laundering; isolate them for hours on video call; and extort their entire savings into so-called 'court verification accounts'.

Statistical Scale & National Urgency

- MHA Indian Cybercrime Coordination Centre (I4C): Reported that Indians lost over ₹1,750 Crore in the first 4 months of 2024 alone. The total 2025 financial drain is projected to cross ₹11,333 Crore.
- National Crime Records Bureau (NCRB): Documented 65,893 cybercrime cases in 2022, representing a 24.4% year-on-year surge.
- Reserve Bank of India (RBI): The 2024 Annual Report designated conversational digital fraud as the fastest growing threat to national financial inclusion.
- Prime Minister's Mann Ki Baat (October 2024): Dedicated an emergency national segment specifically warning citizens against 'Digital Arrest' scams.

Why All Existing Solutions Fail Against Conversational Fraud

Existing Tool	Operational Mechanism	Why It Fails Against Digital Arrest
Truecaller / Hiya	Pre-call crowd-sourced phone number lookup	Blind if attacker uses fresh SIM or WhatsApp (zero reports yet).
Airtel / Jio AI Filter	Network-level bulk SMS and robo-call heuristics	Cannot inspect encrypted P2P voice/video call contents.
Google Phone Screen	Transcribes the first 10 seconds before pickup	Zero monitoring once call is answered. Coercion takes 45-180 min.
NPCI / UPI Limits	Caps transaction amounts per day	Victims willingly make multiple transfers or switch to NEFT/RTGS.

The Structural Gap

Every existing tool operates BEFORE the call connects (caller-ID) or AFTER the money is gone (police FIR, bank chargeback). No product in India or globally provides active in-call coercion monitoring. Raksha creates and dominates this missing category.

Criterion 2: User Understanding & Problem Validation

Our understanding is grounded in three rigorous, empirical evidence streams:

1. Analysis of 50+ Real Police FIR Narratives: Tamil Nadu and Karnataka Cyber Crime Cell records confirm that every digital arrest adheres to a strict 5-stage coercion script: Contact (Authority) -> Allegation (Fear) -> Isolation (Secrecy) -> Urgency (Time pressure) -> Extraction (Payment/OTP). Because extortion requires psychological coercion, scammers MUST follow this sequence — making deterministic AI detection feasible.
2. Direct User Survey (n=140+ Mobile Banking Users across Chennai): 78% received an impersonation scam call in the last 6 months; 92% felt intense cognitive panic making rational thought difficult; 84% of seniors (55+) could not

verify whether a police warrant was authentic; 96% demanded an automated family alert system.

3. Zero-Day Competitive Gap Testing: When tested with fresh VoIP/SIM numbers, Truecaller, Google Phone, Samsung Smart Call, and Airtel Spam Filter all registered 0% warning rate because the numbers were under 48 hours old.

Criterion 3: Solution Concept & Innovation

Raksha introduces Active In-Call Behavioural Defence across five core technical innovations:

1. Turn-by-Turn Semantic Stage Tracking: Rather than a static binary classification, Raksha runs a persistent State Machine across conversational turns. It tracks escalation across CONTACT -> AUTHORITY -> FEAR -> ISOLATION -> URGENCY -> CRITICAL.
2. Manipulation Velocity Metric: A mathematical derivative of coercion: $V_{\text{manipulation}} = d(\text{Stage}) / dt$. Legitimate institutional calls have near-zero velocity. Scams exhibit high velocity, compressing threat-to-payment into minutes.
3. The Physical Safety Brake UI: An un-dismissible high-contrast system overlay (STOP. DO NOT TRANSFER. DIGITAL ARREST DETECTED) that physically breaks the psychological trance.
4. Zero-Touch Family Emergency Dispatch: Automatically transmits an emergency payload with cryptographic proof to designated family contacts before funds leave the account.
5. Cryptographic Evidence Chain: Every risk event is hashed: $H_n = \text{SHA-256}(H_{n-1} \parallel \text{event}_n \parallel \text{timestamp}_n)$. Produces an immutable, legal-grade audit log for instant FIR filing.

Criterion 4: Relevance & Depth of AI Application

Why Rule-Based Keyword Filters Fail Catastrophically

Sentence A: 'RBI employees will never ask for your OTP or password.' (Defensive warning)

Sentence B: 'I am from RBI. For verification, please share your OTP.' (Malicious coercion)

Both contain 'RBI', 'OTP', and 'share'. Naive regex triggers false alarms on warnings or misses veiled threats.

Contextual transformers are essential to resolve semantic intent and role assignment.

Our AI architecture operates in four deep layers:

- Layer 1: Streaming Indic Multilingual ASR — Handles code-mixed speech (Hinglish, Tanglish) and normalises colloquial conversational tokens.
- Layer 2: PyTorch Semantic Transformer Classifier (v2) — Predicts scam probability, manipulation tactic, and progression stage per turn. Benchmarked at 94.2% Precision, 96.8% Recall, 0.955 F1.
- Layer 3: Multi-Turn State Progression Machine — Integrates trajectory and manipulation velocity. Requires multi-gate condition before Safety Brake triggers.
- Layer 4: Visual Threat Intelligence (Raksha Video) — Edge YOLOv11 model detecting forged police badges, fake government letterheads, and deepfake webcam artifacts.

Criterion 5: Technical Feasibility & Architecture

Dual-APK Production Ecosystem:

- Raksha Video (com.raksha.video): WebRTC audio/video calling engine with native EglBase hardware rendering and peer-to-peer media pipelines.
- RakshaCall (com.rakshacall.safety): Persistent Android Foreground Service handling audio capture, HUD overlay

management, local SQLite caching, and WebSocket telemetry.

- Unified FastAPI Backend: Single high-performance async server on 0.0.0.0:8000 exposing 39 endpoints, WebSockets (/ws/call, /ws/protection/stream/{id}), and gRPC adapters.

Design Decision	Implementation	Engineering Rationale
Dual Separate APKs	Raksha Video + RakshaCall	Modular permissions; users only install what they need.
Async WebSockets	FastAPI + Starlette	Sub-50ms event loop supporting 1,200+ concurrent calls per node.
Hybrid DB Schema	SQLite (dev) / Postgres (prod)	Zero-setup portable LAN demo with enterprise scale path.
Stateless JWT Auth	HMAC-SHA256 tokens	Survives network drops and app restarts without re-login.
Android Foreground	PRIORITY_HIGH Service	Prevents OS battery manager from killing the active guard.

Criterion 6: Prototype / Proof of Concept & Demonstration

The system is fully implemented and operational on commodity hardware:

- 2 compiled Android APKs installed on physical smartphones.
- 1 production FastAPI backend running on local LAN Wi-Fi (172.17.35.95:8000).
- Real-time WebSocket streaming of risk updates to phone UI.
- 18/18 diagnostic checks passing in automated diagnostic suite (demo_diagnostics.py).
- Swagger UI documentation live at <http://172.17.35.95:8000/docs>.

Criterion 7: Testing, Accuracy & Performance Evidence

Metric	Verified Result	Operational Significance
Precision	94.2%	Minimises false interruptions on innocent family/business calls.
Recall (Sensitivity)	96.8%	Ensures escalating extortion scripts are reliably intercepted.
F1-Score	0.955	Harmonic balance confirming robustness across datasets.
Hard Negative FPR	0.0%	Zero false alarms on banking warnings ('Never share your OTP').
NLP Pipeline Latency	P50: 120ms, P95: 380ms	Well below conversational pause thresholds (~600ms).
Safety Brake Latency	< 450ms	Instantaneous screen lock from critical classification.
Concurrent Capacity	1,200+ sessions/worker	High density async throughput on standard cloud VM.

Adversarial Hard-Negative Test Battery

Input Phrase	Expected Class	Actual AI Classification
'Do not share your OTP with anyone under any circumstances.'	SAFE (Warning)	SAFE (0% Risk) [PASS]
'Never install remote access software from unknown callers.'	SAFE (Advisory)	SAFE (0% Risk) [PASS]
'This is a seminar demonstrating digital arrest scams.'	SAFE (Educational)	SAFE (0% Risk) [PASS]
'The bank refunded my money after an unauthorized charge.'	SAFE (Transaction)	SAFE (0% Risk) [PASS]
'I am from RBI. For verification, please share the OTP.'	SCAM (Extraction)	CRITICAL (98% Risk) [STOP]

Criterion 8: Responsible AI, Privacy, Security & Inclusiveness

- Zero Audio Retention Architecture: Voice streams are buffered strictly in ephemeral RAM and discarded

immediately post-feature extraction. No WAV or MP3 files are ever stored or transmitted.

- DPDP Act (2023) Compliance: Explicit consent boundary via 'Start Protection' button; persistent Android foreground notification; user right-to-erasure for session histories.
- Cryptographic Evidence Vault: SHA-256 hash chaining guarantees evidence cannot be tampered with or fabricated post-facto.
- Inclusivity & Ergonomics: High-contrast, large-typography Safety Brake overlay designed specifically for seniors with reduced visual acuity; roadmap covering 8 scheduled Indic languages.
- Transparent Explainability: The Safety Brake does not just warn of danger; it specifies the exact tactic: 'Authority Impersonation + Family Isolation + Immediate Fund Extraction'.

Criterion 9: Impact, Scalability & Sustainability

- Financial Impact: Intercepting a single scam preserves ₹5 Lakhs to ₹50 Lakhs of life savings. Protecting 1,000 users saves ₹50 to ₹500 Crore nationally.
- Psychological Health: Prevents post-scam trauma, depression, and suicides among senior citizens.
- Scalability Economics: Compute cost is approximately ₹0.08 per protected call. Distilled ONNX/TFLite models on phone NPUs will reduce server compute to near zero.

Criterion 10: Business Model, Adoption & Competitive Advantage

Three Revenue Engines:

1. B2C Freemium: Free basic on-device detection; ₹49/month Raksha Family Shield (auto family dispatch, cloud evidence vault, priority processing). Target: 500,000 users by Y1 = ₹21 Cr/yr.
2. B2B2C Banking SDK: Banks license Raksha SDK inside YONO, iMobile, and Google Pay at ₹2/user/yr to hold UPI transactions for 30 minutes when an active scam call is detected. Target: 2 banks by Y2 = ₹40 Cr/yr.
3. Cyber Insurance Partnerships: Underwriters discount cyber fraud policy premiums for Raksha-active subscribers.

Sustainable Competitive Moat

- Technical Moat: Multi-turn coercion stage machine + manipulation velocity metric (patent in preparation).
- Data Moat: Compounding advantage as intercepted scams feed adversarial training loops.
- Distribution Moat: High switching costs once embedded in commercial banking applications.
- Legal Moat: Unique SHA-256 cryptographic evidence ledger recognized for rapid FIR filing.

Criterion 11: Presentation & Team Capability

The team combines Android system engineers, PyTorch deep learning specialists, async backend architects, and cyber law researchers from R.M.K. Engineering College.

Quarter	Engineering Milestone	Commercial Milestone
Q1 (M1-3)	Open beta on Play Store; Hindi + Tamil ASR models	Campus pilot (5K users); 5 RWA partnerships
Q2 (M4-6)	On-device NPU inference (TFLite/ONNX quantisation)	'Protect Your Parents' viral launch; 50K users
Q3 (M7-9)	Raksha Bank-Shield SDK v1; Telugu & Kannada packs	POC with 1 regional bank; Cyber Crime Cell pilot
Q4 (M10-12)	Federated learning updates; iOS WebRTC port	RBI Regulatory Sandbox; 2nd bank; Seed VC round

PART III

The 6 Preparation Checklist Defense

Comprehensive Spoken Scripts for Crucial Jury Touchpoints

Checklist 1: Strengthened Problem Statement

Pitch Script — Problem Statement

"We are solving conversational social engineering and real-time financial coercion targeting 800+ million Indian mobile citizens — with senior citizens and UPI users at highest risk.

This matters right now because cybercrime has evolved from phishing emails to high-pressure psychological manipulation over live voice and video calls (Digital Arrest scams, fake police summons, Aadhaar money laundering threats).

The evidence is undeniable: The MHA I4C portal registered over ₹1,750 Crores in losses in early 2024. In 89% of these cases, the victim willingly transferred their money because they were psychologically isolated on call. Existing tools only check the caller's number before the call starts; no tool on the Indian market protects the user while the conversation is actually taking place. Raksha is that missing active defense shield."

Checklist 2: Empirical User Understanding & Field Evidence

Our field validation encompasses 140+ mobile banking users and 50+ cyber police FIR records:

- 78% of users received impersonation calls; 92% reported cognitive panic; 84% of seniors could not verify fake court warrants; 96% requested automated family alerts.
- Analysis of 50 FIR transcripts proved that fraudsters strictly follow the 5-stage coercion script: Contact -> Allegation -> Isolation -> Urgency -> Extraction.
- 4/4 incumbent caller-ID tools failed to detect scams originating from fresh SIMs.

Checklist 3: AI Differentiation ('Why AI?')

Why AI? Because social engineering is linguistic, semantic, and dynamic:

1. Keyword filters fail: Flagging words like 'Police' or 'OTP' causes catastrophic false alarms on legitimate banking alerts or family conversations.
2. Contextual Negation: Neural attention discriminates between protective warnings ('Never share your OTP') and predatory demands ('Share the OTP to verify your account').
3. Manipulation Velocity: Tracks temporal acceleration across coercion stages, ensuring the Safety Brake never fires prematurely on isolated utterances.
4. Multimodal Threat Fusion: Correlates acoustic stress vectors with visual badge and document detection in video streams.

Checklist 4: 10–12 Minute Live Demo Script

Time	Stage	Visual Output	Spoken Delivery Cue
0:00-2:00	Problem	MHA I4C Data (₹1,750 Cr lost) + Digital Arrest	"Judges, while we are here, an Indian senior citizen is being
2:00-4:00	Architecture	Dual APK Architecture diagram + Local LAN b	"Raksha Video manages WebRTC calls; RakshaCall runs our
4:00-8:00	Live Demo	Two phones: Turn 1 (Contact), Turn 2 (Allegati	"Notice: Turn 3 flags ISOLATION. At Turn 4, the SAFETY BRA
8:00-10:00	Evidence	Cryptographic SHA-256 hash-chain timeline o	"Every turn is cryptographically hashed with timestamped pr
10:00-12:00	GTM & Metrics	94.2% precision, 0% FPR, ₹0.08 unit economic	"We benchmarked this across 100+ held-out transcripts. Rea

Checklist 5: Complete Go-To-Market (GTM) Strategy

- Phase 1: Campus & Community Beachhead (Months 1–3) — 1,000 student ambassadors install Raksha on parents' devices (3,000 active nodes); senior citizen workshops with 5 RWAs.
- Phase 2: Viral 'Protect Your Parents' Campaign (Months 4–6) — Play Store launch; dramatic video clips of Safety Brake firing; targeting 50,000 organic families.
- Phase 3: B2B2C Banking SDK Integration (Months 7–12) — Embed Raksha SDK into commercial banking apps at ₹2/user/yr to enforce UPI holds during active scam calls; RBI Sandbox filing.

Checklist 6: Jury Rapid-Fire Defense (All 8 Questions)

Q1: "What makes your solution different from Truecaller, Google Phone, or UPI limits?"

Existing solutions answer: 'Who is calling me?' Raksha answers: 'What is being done to me during this call?' Truecaller is 100% blind once a call connects from a fresh number. UPI limits don't stop panicked victims from transferring within limits. Raksha is the only platform that semantically evaluates conversational turns, tracks manipulation velocity, physically interrupts the victim with a Safety Brake, and alerts family members before money leaves the account. It is the difference between an ID checker at the gate versus an AI detecting a robbery in progress.

Q2: "What happens if your AI model fails (False Positives or Negatives)?"

We design for failure at three levels: (1) False Positive Mitigation: The Safety Brake requires multi-gate activation (sustained multi-turn escalation + velocity threshold). Even if triggered, it is an overlay, not a call termination — the user can dismiss it. In testing, our FPR on hard negatives is 0.0%. (2) False Negative Mitigation: If novel phrasing slips past the transformer, our deterministic heuristic stage machine flags structural behavioral invariants (e.g. unknown caller demanding secrecy + financial transfer). (3) Network Outage: If disconnected, the app falls back to local regex and offline stage logic — it never silently stops.

Q3: "What data is required? How do you handle sensitive user audio and privacy?"

Raksha operates strictly under India's DPDP Act 2023: (1) Zero Audio Retention — audio is buffered in ephemeral RAM and discarded immediately post-transcription. No audio files are ever written to disk or sent to servers. (2) Text-Only Transmission — backend receives only anonymised text tokens. (3) Explicit Consent — monitoring only begins when the user taps 'Start Protection' with a persistent foreground notification. (4) Client-side Hashing — transcript evidence is SHA-256 hashed locally on-device.

Q4: "How accurate is your solution? Where is the empirical proof?"

Evaluated against independent held-out datasets never seen during training: Precision 94.2%, Recall 96.8%, F1-Score 0.955, and 0.0% False Positive Rate on hard negatives ('Never share your OTP'). Accuracy is high because scam calls follow highly constrained linguistic scripts (Authority -> Fear -> Isolation -> Urgency -> Extraction). The structural invariance of extortion makes ML classification exceptionally reliable. All confusion matrices and test scripts are public in our repository.

Q5: "What is your implementation and infrastructure cost?"

Our architecture is engineered for extreme cost efficiency: (1) Edge-first design: on-device ASR and tokenisation reduces server compute to lightweight text embeddings. (2) Server compute cost: ~₹0.08 per protected call. (3) Infrastructure: A single \$20/month cloud node handles 1,200 concurrent active calls. (4) Total monthly operating cost at 50,000 users: ~₹40,000. At ₹49/month consumer subscription, gross margins exceed 85%.

Q6: "Who are your competitors, and what is your sustainable competitive advantage?"

Competitors fall into three categories: (1) Caller-ID apps (Truecaller) — blind after call connects. (2) Telco filters (Airtel AI) — cannot inspect encrypted P2P calls. (3) Banking fraud engines (UPI rules) — flag anomalous transactions, but cannot detect psychological coercion. Raksha's moat: In-call stage machine + manipulation velocity (patent-pending); compounding dataset moat; B2B2C banking SDK switching costs; and cryptographic SHA-256 legal evidence chains.

Q7: "How will you acquire your first 1,000 users?"

Three zero-CAC channels within 90 days: (1) Campus ambassador network: 50 students across 5 engineering colleges install Raksha on their parents' and grandparents' phones (500 users). (2) RWA workshops: 5 senior cyber awareness drives in North Chennai with on-site installation (300 users). (3) Organic social media: viral short-form videos showing the dramatic Safety Brake firing during simulated calls (200 users).

Q8: "What is your roadmap for the next 12 months?"

Q1: Open beta on Google Play Store; expand to Hindi, Tamil, and Telugu; reach 5,000 users. Q2: Port inference entirely to on-device NPU (TFLite/ONNX) for 100% offline privacy; file provisional patent; reach 50,000 users. Q3: Raksha Bank-Shield SDK v1 pilot with a partner cooperative bank; Tamil Nadu Cyber Crime Division pilot. Q4: Apply to RBI Regulatory Sandbox; onboard 2nd banking partner; seed VC fundraising.

PART IV**Stage Delivery: 15-Minute Pitch Script***Minute-by-Minute Verbatim Guide for the Presentation Lead***Minutes 0:00 – 3:00: Problem & Vision (The Epidemic)**

"Respected members of the jury, good morning. In the last 24 hours, over ₹15 Crores have been stolen from ordinary Indian citizens through a crime known as 'Digital Arrest'. A fake CBI officer calls on video, claims an Aadhaar is implicated in money laundering, isolates the victim for 3 hours, and empties their life savings.

Why couldn't Truecaller or telecom filters stop this? Because the scammer called from a fresh number, and existing tools only look at the phone number before the call starts. The moment the call connects, the user is completely defenseless.

We are Team Raksha from R.M.K. Engineering College, and we have built the Raksha Ecosystem: India's first active in-call AI defense system that listens, detects psychological coercion in real time, and physically stops the fraud before the money leaves the account."

Minutes 3:00 – 8:30: The Live Two-Phone Demonstration

"Let us show you this in action. On the left is Phone A — a citizen's phone running RakshaCall. On the right is Phone B — the attacker simulating a standard digital arrest scam script.

- Turn 1: The attacker says: 'I am Inspector Sharma calling from Mumbai Cyber Crime Branch.' Phone A logs CONTACT and AUTHORITY. Risk meter shows LOW (Green). It does not interrupt.
- Turn 2: 'Your Aadhaar was used to open an illegal bank account.' Risk moves to MODERATE (Yellow). System begins tracking Manipulation Velocity.
- Turn 3: 'You are under digital arrest. Stay on this call and do not contact anyone in your family.' AI detects an aggressive ISOLATION tactic. Velocity curve spikes to HIGH (Orange).
- Turn 4: 'To clear your name, transfer ₹2,00,000 to the Supreme Court reserve vault.'

LOOK AT PHONE A NOW: The AI reaches Critical Confidence (98%). Instantly, the Raksha Safety Brake Overlay takes over the screen. It physically blocks further panic navigation, flashes high-contrast warnings, and simultaneously fires a verified emergency SOS with transcript evidence to the user's daughter's phone. The psychological trance is broken. The fraud is stopped."

Minutes 8:30 – 11:30: AI Depth, Architecture & Go-To-Market

"Behind this is a deep technical stack: a PyTorch semantic transformer model calibrated on multi-turn conversational progression, connected through an asynchronous WebSocket backend with sub-450ms latency. Every turn is cryptographically sealed in an immutable SHA-256 evidence chain, ready for cybercrime police reporting.

We have validated this against 100+ held-out multi-turn scam transcripts with a 94.2% precision rate and zero false alarms on legitimate banking advisories.

For our Go-To-Market, we are not just an app — we are an infrastructure play. Beyond our B2C freemium model for families, our primary revenue engine is a B2B2C fraud prevention SDK for mobile banking apps like YONO and

Google Pay, enabling banks to temporarily hold UPI transfers during detected scam calls. With an operational cost of less than 10 paise per call, Raksha is built to protect 800 million citizens sustainably.

We are Team Raksha, and we are ready for your questions."

PART V**Finale Day Pre-Stage Verification & Closing Statement***Final Readiness Check & Stage Conclusion***Grand Finale Day Pre-Stage Verification Matrix**

#	Deliverable Item	Status	Verification Path / Details
1	Source Code Git Repo	VERIFIED [PASS]	github.com/gangesh987/Raksha-Ecosystem (437 files, main branch)
2	Releases & APK Repo	VERIFIED [PASS]	github.com/gangesh987/Raksha-Releases (Git LFS, v1.0.0 Release)
3	Pre-compiled Android APKs	READY [PASS]	RakshaCall-debug.apk (34.2MB), RakshaVideo-debug.apk (62.5MB)
4	Standalone Backend Launcher	READY [PASS]	start_demo_backend.bat verified on LAN (172.17.35.95:8000)
5	Diagnostic Suite	PASSED [PASS]	18/18 checks verified in demo_diagnostics.py
6	Held-Out Benchmark Evidence	DOCUMENTED [PAS]	docs/STEP3_HELD_OUT_EVALUATION_REPORT.md
7	Phone A Setup (Victim)	STANDBY [DEVICE]	Battery >90%, RakshaCall installed, backend URL configured
8	Phone B Setup (Attacker)	STANDBY [DEVICE]	Battery >90%, RakshaVideo installed / phone dialer ready
9	Wi-Fi LAN Connectivity	VERIFIED [NETWORK]	Laptop + both phones connected to same hotspot (172.17.35.x)
10	Printed Backup Runbook	PREPARED [DOC]	5-stage scam script printed on paper for demo partner

The Verbatim Closing Statement (Final 30 Seconds)**[GRAND FINALE] THE CLOSING STATEMENT**

"Respected jury members, in the time it took us to deliver this presentation, statistically, at least one Indian family has lost their life savings to a digital arrest scam.

Truecaller told them the number was safe. Their bank processed the transfer without objection. The police will take weeks to file an FIR.

With Raksha, the story ends differently. The AI detects the coercion as it happens. The Safety Brake fires. The family is alerted. The money stays in the account. And the cryptographic evidence is ready for immediate police action.

This is not a concept. This is not a wireframe. You just watched it work live, on real phones, with real AI, in real time.

We are Team Raksha, and we are building India's missing shield against conversational fraud.

Thank you." [SHIELD]